

Java Stream API — `filter()` + `map()` Practice

1. Even numbers ka square

```
List<Integer> numbers =  
    List.of(10, 15, 20, 25, 30, 35, 40);
```

`filter()` aur `map()` ka use karke **sirf even numbers ka square** print karo.

Expected:

```
100  
400  
900  
1600
```

2. Odd numbers ko double karo

```
List<Integer> numbers =  
    List.of(5, 10, 15, 20, 25, 30);
```

Sirf odd numbers filter karo aur unko `2` se multiply karo.

Expected:

```
10  
30  
50
```

3. Greater than 20 → multiply by 10

```
List<Integer> numbers =  
    List.of(10, 20, 30, 40, 50);
```

20 se greater numbers filter karo aur har number ko `10` se multiply karo.

4. Names starting with A → uppercase

```
List<String> names =  
    List.of("Amit", "Rahul", "Anil", "Rohit", "Ajay");
```

Sirf "A" se start hone wale names filter karo aur uppercase me convert karo.

Expected:

AMIT
ANIL
AJAY

5. Name length > 4 → uppercase

```
List<String> names =  
    List.of("Amit", "Rahul", "Anil", "Suresh", "Raj");
```

Jinki length 4 se greater hai un names ko filter karo aur uppercase karo.

6. Positive numbers → square

```
List<Integer> numbers =  
    List.of(-5, 10, -20, 15, 30, -40);
```

Sirf positive numbers filter karo aur unka square print karo.

7. Salary > 30,000 → 10% increment

```
List<Integer> salaries =  
    List.of(25000, 35000, 45000, 28000, 60000);
```

₹30,000 se greater salary wale employees ko filter karo aur salary me **10% increment** apply karo.

Example:

35000 → 38500
45000 → 49500
60000 → 66000

8. Even numbers → add 100

```
List<Integer> numbers =  
    List.of(10, 11, 12, 13, 14, 15);
```

Even numbers filter karo aur har number me 100 add karo.

Expected:

110
112
114

9. Names containing "a" → uppercase

```
List<String> names =  
    List.of("Amit", "Rahul", "Rohit", "Suresh", "Anil");
```

Jin names me "a" present hai unhe filter karo aur uppercase me convert karo.

Hint: `contains()` use kar sakte ho.

10. Real-world Students

```
List<Integer> marks =  
    List.of(35, 45, 60, 72, 28, 90, 55, 40);
```

Stream API ka use karke:

1. 40 ya usse greater marks filter karo
2. Har selected mark me 5 grace marks add karo
3. Result print karo

Expected:

50
65
77
95
60
45